SEDATIVE AND HYPNOTICS

Learning Objectives

1. Differentiate sedation, hypnosis, anesthesia, and coma based on their clinical features and depth of CNS depression.

2. Describe the structure, subunit heterogeneity, and physiological functions of the GABAA receptor-chloride channel complex and explain their therapeutic implications.

3. Explain the mechanisms of action of sedative, hypnotic, and anxiolytic drugs that enhance GABA-A receptor function.

4. Identify sedative, hypnotic, and anxiolytic drugs that act independently of GABA-A enhancement and describe their molecular targets.

5. Recognize the clinical manifestations of benzodiazepine and barbiturate overdose and explain their management, including the mechanism, indications, and limitations of flumazenil.

6. Compare the dependence potential, tolerance, and withdrawal syndromes associated with different classes of sedatives, hypnotics, and anxiolytics.

Introduction to CNS Pharm

‘Neurotransmitters’

- Neurotransmitters are substances which neurons use to communicate with one another and with their target tissues in the process of synaptic transmission (neurotransmission).

- Neurotransmitters are synthetized in and released from nerve endings into the synaptic cleft.

- From there, neurotransmitters bind to receptor proteins in the cellular membrane of the target tissue.

- The target tissue gets excited, inhibited, or functionally modified in some other way.

- There are more than 40 neurotransmitters in the human nervous system.

Introduction to CNS Pharm

Excitatory neurotransmitters	Glutamate (Glu) Acetylcholine (ACh) Histamine Dopamine (DA) Norepinephrine (NE); also known as noradrenaline (NAd) Epinephrine (Epi); also known as adrenaline (Ad)
Inhibitory neurotransmitters	*gamma*-Aminobutyric acid (GABA) Serotonin (5-HT) Dopamine (DA)
Neuromodulators	Dopamine (DA) Serotonin (5-HT) Acetylcholine (ACh) Histamine Norepinephrine (NE)
Neurohormones	Releasing hormones from hypothalamus Oxytocin (Oxt) Vasopressin; also known as antidiuretic hormone (ADH)

Introduction to CNS Pharm

Disorders associated with neurotransmitters

Alzheimer's disease

Alzheimer's disease is a neurodegenerative disorder characterized by learning and memory impairments. It is associated with a lack of acetylcholine in certain regions of the brain.

Depression

Depression is believed to be caused by a depletion of norepinephrine, serotonin, and dopamine in the central nervous system.

Schizophrenia

Schizophrenia, which is a severe mental illness, has been shown to involve excessive amounts of dopamine in the frontal lobes, which leads to psychotic episodes in these patients.

Introduction to CNS Pharm

Disorders associated with neurotransmitters

Parkinson's disease

The destruction of the substantia nigra leads to the destruction of the only central nervous system source of dopamine.

Epilepsy

Some epileptic conditions are caused by the lack of inhibitory neurotransmitters, such as GABA, or by the increase of excitatory neurotransmitters, such is glutamate.

Huntington's disease

Besides epilepsy, a chronic reduction of GABA in the brain can lead to Huntington's disease.

Sedation- decrease in alertness and response to any stimulation without inducing sleep

Hypnosis- a state that resembles natural sleep but the person can be aroused by strong stimuli like pin prick or the sound of the alarm clock

Anxiety- a term for a disorder which is characterized by symptoms such as tension, fear, lack of concentration, tachycardia, tremors, sweating

SEDATIVES

- Drugs that reduces excitement, calms the subject without inducing sleep
- In therapeutic doses, are anxiolytic agents
- Larger doses produce hypnosis
- Site of action- limbic system (that regulates thought and mental function)

HYPNOTICS

- Drug that induce sleep resembling natural sleep
- Used for initiation and maintenance of sleep
- In higher doses produce general anaesthesia
- Site of action- midbrain and ascending reticular formation(which maintains wakefulness)

Different phases of sleep

- **Stage 0(awake)-** 1-2% of sleep time

Lying down to falling asleep
Eye movement- irregular or slowly moving

- **Stage 1(dozing):** 3-6% of sleep time

Eye movement- reduced
Neck muscles- relaxed

- **Stage 2(unequivocal sleep)-40 to 50% of sleep time**

Eye movement- little
Subjects easily arousable

- **Stage 3(deep sleep transition)- 5 to 8% of sleep time**

Eye movement- very little
Subject not easily arousable

Stage 4(cerebral sleep)- 10-20% of sleep time

- ✔ Deepest level of sleep
- ✔ Metabolic rate is less
- ✔ Growth hormone secretion is high
- ✔ Eye movement- fixed
- ✔ Subjects difficult to arouse
- ✔ Night terror occur at this time

- **REM sleep(Rapid Eye Movement)/ paradoxical sleep**

Comprises 20-30% of the sleep time

Eye movement- marked, irregular & rapid eye movement

Dreams and nightmare occurs which can be recalled

Heart rate and BP fluctuates

Respiration- irregular

Muscles- fully relaxed

Erection- in case of males

- Stages 0-4 and REM sleep- 80 to 100 mins
- Stage 1-4- REM sleep repeated cyclically

Classification

- **Benzodiazepines-**

Hypnotic- Diazepam, Flurazepam, Nitrazepam, Alprazolam, Lorazepam, Temazepam, Triazolam

Antianxiety- Diazepam, chlordiazepoxide, Oxazepam, Lorazepam

Anti-convulsant- Clonazepam, Lorazepam, Diazepam

- **Non- Benzodiazepine Hypnotics-**

Zolpidem, Zaleplon, Zopiclone, Eszopiclone

- **Barbiturates-**

Long acting- Phenobarbitone

Short acting- Butobarbitone, Pentobarbital

Ultrashort acting- Thiopental, Methohexital

- **Atypical Anxiolytics-**

Buspirone, Ipsapirone, Gepirone

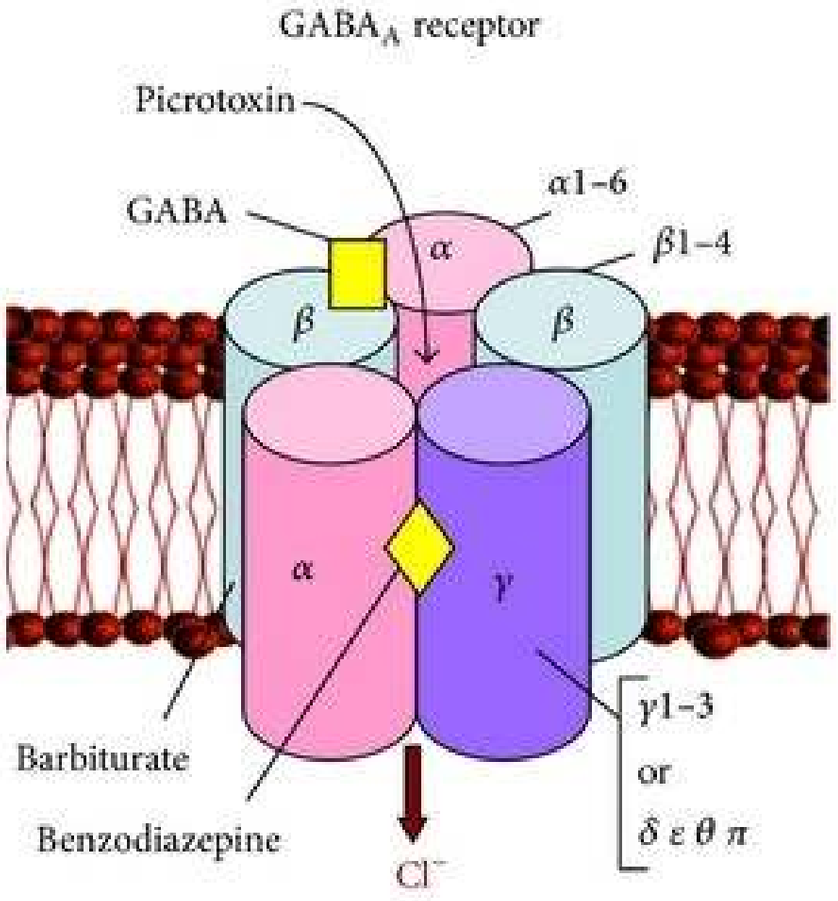
- **Beta-adrenoceptor antagonists-**

Propranolol

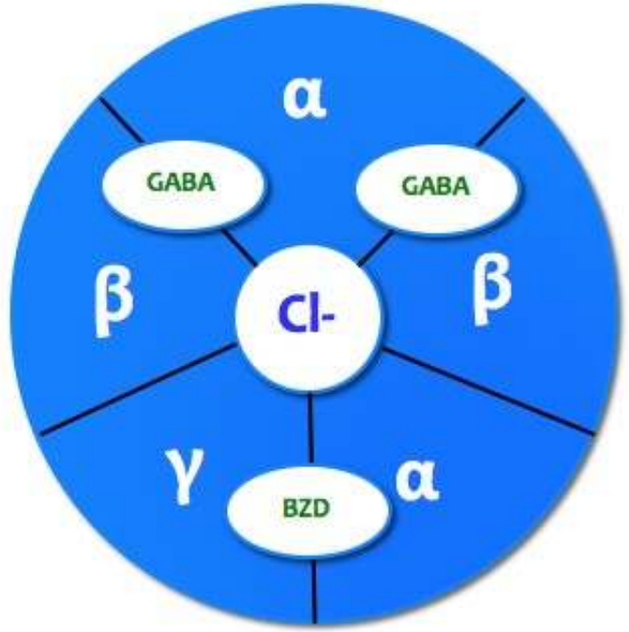
- **Miscellaneous-**

Melatonin, Ramelteon

GABA Receptor Cl^- Channel Complex



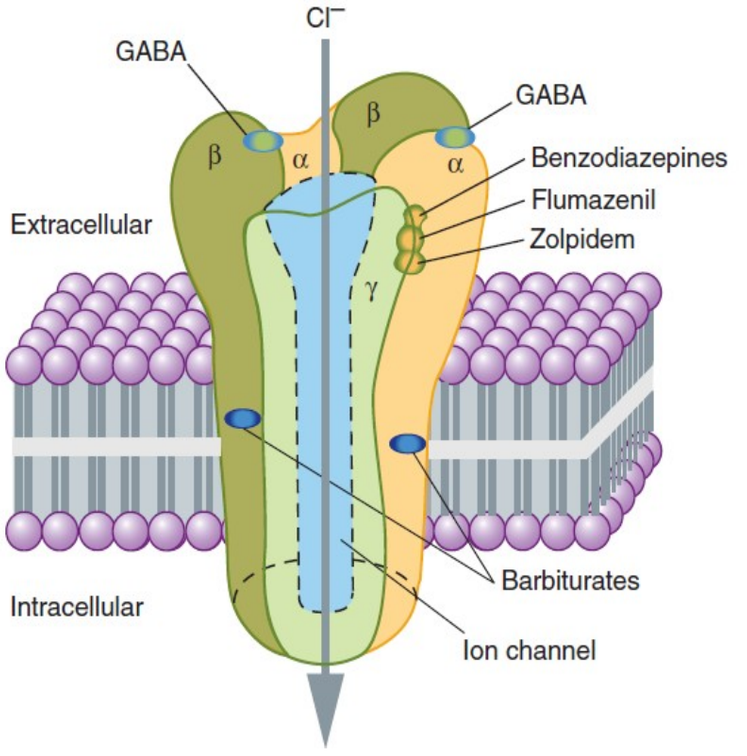
GABA = major inhibitory neurotransmitter in the CNS



Top down view

Barbiturates

- Acts primarily at GABA: BZD receptor chloride channel complex
- Potentiates the GABA mediated inhibitory effects by increasing the duration of chloride channel opening
- At higher doses, it directly increase the chloride ion conductance(GABA mimetic action)



Pharmacokinetics

- Rate of absorption depends on lipid solubility
- Given slow I.V, not suitable for I.M/S.C routes
- Can be given orally and rectally
- Widely distributed depending upon their lipid solubility and regional blood flow to different parts of the body
- Metabolised by both phase 1 and phase 2 processes
- Excreted through urine

Pharmacological actions

1. Central nervous system

- ❖ Sedation and hypnosis
- ✔ Induces sleep
- ✔ Prolong duration of sleep
- ❖ Alters NREM and REM sleep cycle
- ❖ Anaesthesia
- ❖ Anticonvulsant

2. Respiratory system

- ❖ Significant respiratory depression

3. CVS

Higher doses-decrease BP and HR and depresses myocardium

4. GIT

Decrease in tone and amplitude of GIT smooth muscles

5. LIVER

Enzyme induction- drug interactions

Therapeutic uses

- In anaesthesia- ultrashort acting (thiopental) as inducing agent
- As anticonvulsants- long acting(phenobarbital)
- To treat hyperbilirubinaemia of neonates

Increases the activity of glucuronyl transferase- increase the conjugation of bilirubin- decrease the risk of Kernicterus

Contraindications

- Liver dysfunction
- kidney disease
- Severe pulmonary insufficiency
- Acute intermittent porphyria-

Barbiturates causes induction of ALA-synthetase enzyme in mitochondria- leads to increased porphyrins

Acute barbiturate poisoning

- Mostly suicidal, sometimes accidental
- Patient is flabby and comatose with shallow and failing respiration, fall in B.P.

Treatment-

- No specific antidote
- Maintenance of patient’s vitals
- Gastric lavage- activated charcoal
- Alkaline diuresis- sodium bicarbonate
- Hemodialysis and hemoperfusion

Drug Interactions:

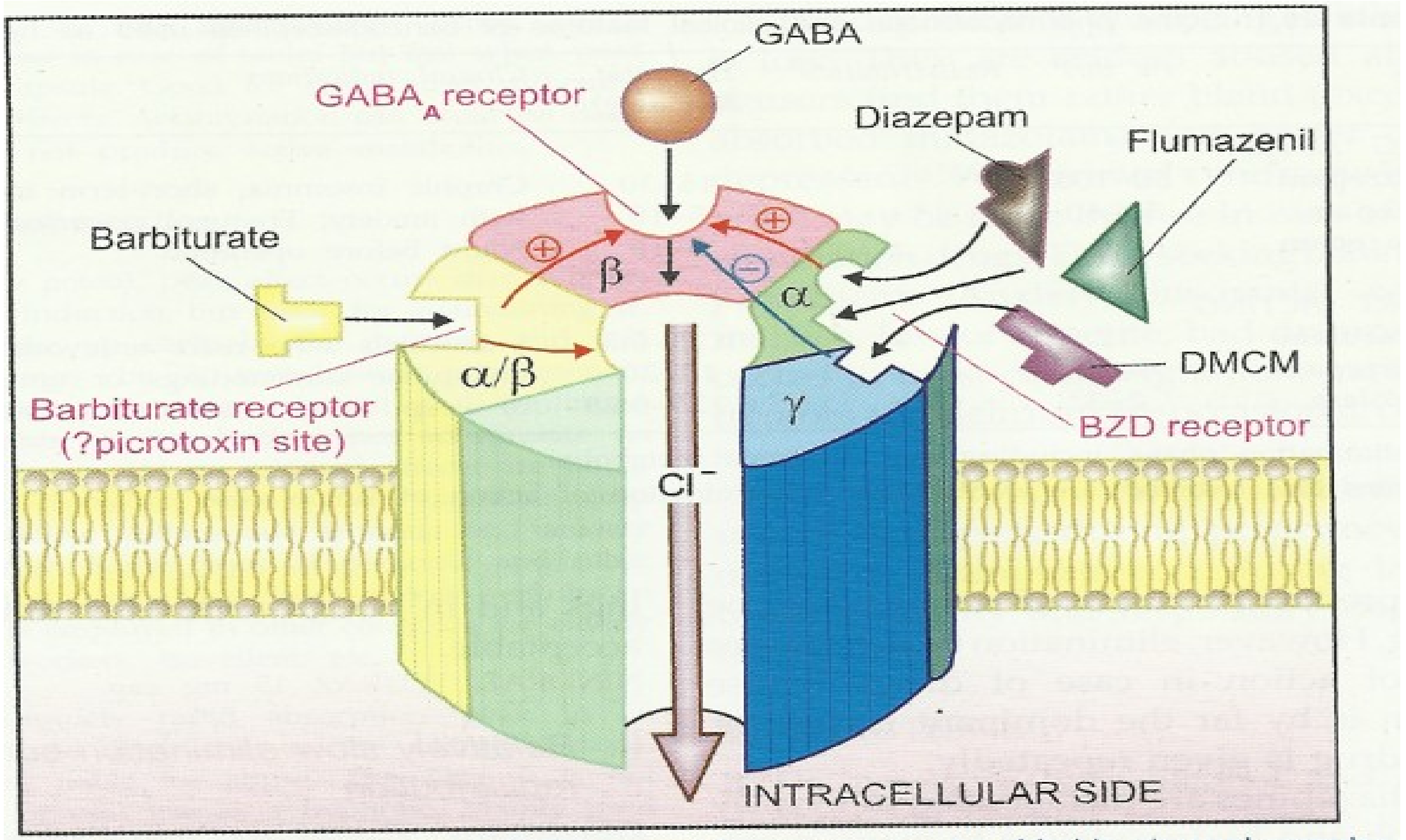
- Induce metabolism of many drugs – warfarin, steroids, OCP, chloramphenicol, tolbutamide
- Alcohol, antihistamines, opioids – CNS depression
- Phenobarbitone reduces the absorption of griesofulvin from gut

Benzodiazepines

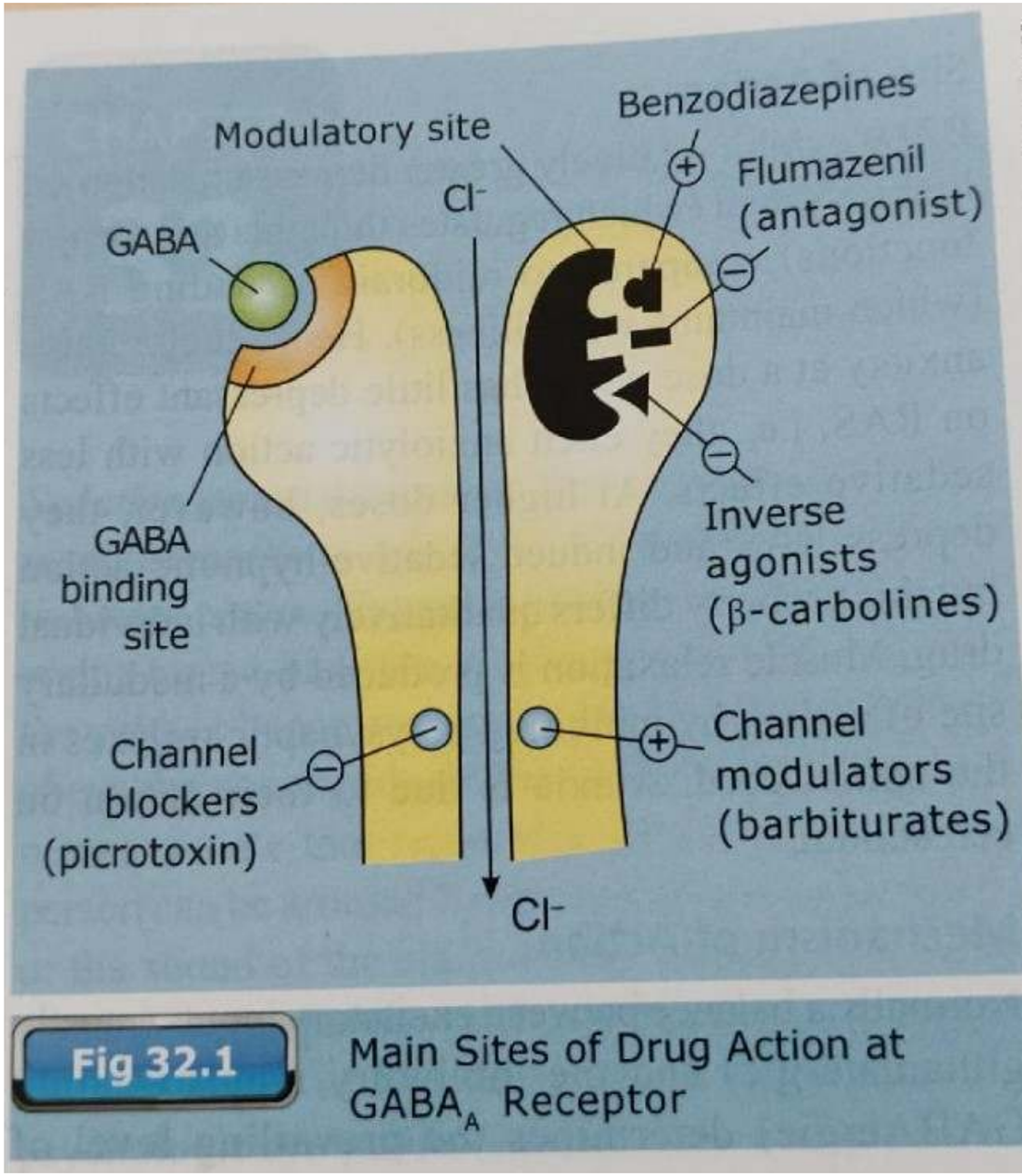
- Chlordiazepoxide- first benzodiazepines discovered by Leo Sternbach in 1955

- BZDs enhances the effect of neurotransmitter GABA-A which is an inhibitory resulting in sedative, hypnotic, anxiolytic, anticonvulsant and muscle relaxant properties

MOA-


GABA
GABA$_A$ receptor
Diazepam
Flumazenil
Barbiturate
β
α
α/β
DMCM
γ
Barbiturate receptor
(?picrotoxin site)
BZD receptor
Cl$^-$
INTRACELLULAR SIDE

- Benzodiazepines
- Binds specific site at GABA A-BZD receptor
- Enhances receptor affinity for GABA
- Increases the frequency of opening of Cl channel
- Increase chloride conduction
- Neuronal membrane hyperpolarization
- Decrease sympathetic transmission
- CNS depression



Pharmacokinetics

- Marked pharmacokinetic differences
- Except midazolam all can be given orally
- Plasma protein binding also varies (Flurazepam – 10%; Diazepam 99%)
- Widely distributed in body
- Metabolised in liver mainly by CYP3A4 and CYP2C19
- Excreted in urine
- Crosses BBB- contraindicated in pregnancy

Therapeutic uses

- To treat anxiety
- To treat insomnia
- As a preanaesthetic medication
- As a skeletal muscle relaxants
- Anti convulsant
- To treat alcohol withdrawal

Adverse effects

- Dose dependent drowsiness, fatigue, disorientation, lethargy and psychomotor skills impairment
- Fast I.V injection- cardiac arrest
- BZD withdrawal includes- anxiety, insomnia, impaired concentration, headache, irritability, tremors, palpitaion and vivid dreams

Flunitrazepam- has sedative amnesic effects

Misused in case of sexual assaults- to wipe out memory of events

Drug interaction

- BZDs potentiates the effects of other CNS depressants like alcohol, hypnotics and neuroleptics
- Smoking decreases the activity of BZDs
- Aminophylline antagonises sedative effects of BZDs
- Enzyme inhibitors like cimetidine and ketoconazole enhance BZD action

Benzodiazepine antagonists

Flumazenil

- Selective competitive antagonists of Benzodiazepine
- Competes with BZD agonist and inverse agonist- reverses their action
- Injected I.V
- Action starts in seconds and lasts for 1-2 hours
- Elimination half life – 1 hour

Uses

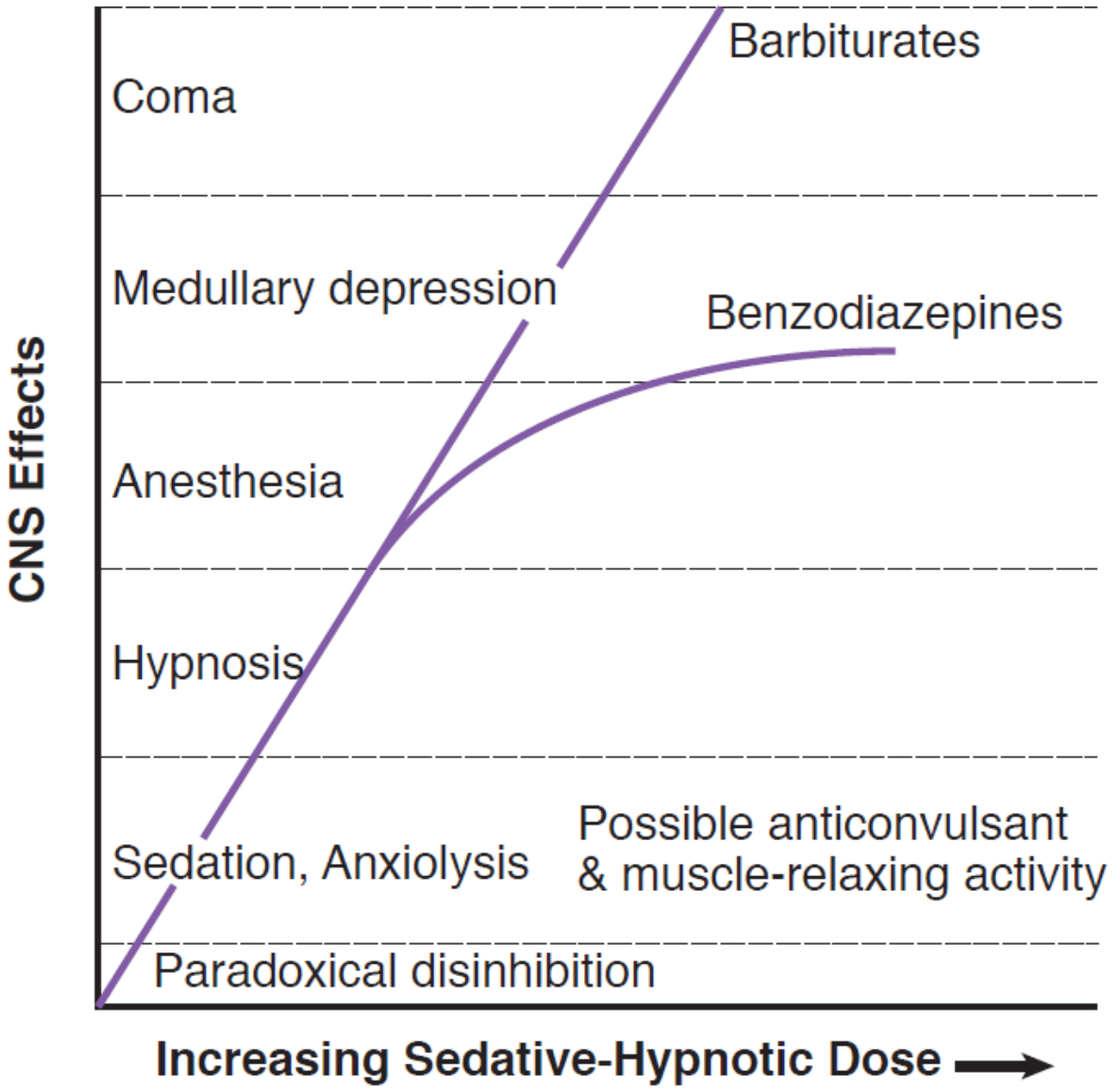
- To reverse BZD anaesthesia
- BZD overdose

Benzodiazepines over Barbiturates as sedatives and hypnotics

- BZDs have **flat DRC** whereas barbiturates have a **steep DRC**
- BZDs does not affect REM sleep and hence cause less hangover whereas Barbiturates causes marked suppression of REM sleep, hence rebound increase in REM sleep on withdrawal
- BZDs **are not enzyme inducers**, less likely to cause drug interactions whereas Barbiturates are **potent enzyme inducers**

- BZDs has low abuse liability whereas barbiturates exhibit tolerance as well as physical and psychological dependence
- BZDs causes amnesia with no phenomenon of **automatism** while barbiturates are characterized by amnesia with automatism
- BZDs shows no hyperalgesia while it is seen with barbiturates
- BZDs poisoning can be treated with specific antagonists **Flumazenil** while no antidote is available for Barbiturate poisoning

Dose Dependent CNS Depression



Insomnia

- a common condition characterized by difficulty falling asleep, increased nighttime wakefulness or inadequate sleep duration
- results in daytime consequences like-

- tiredness
- Lack of concentration
- Irritability
- Reduced work productivity
- Lower quality of life
- Decrements in mood along with memory or cognitive functioning.

Transient insomnia	Short-term insomnia	Long-term insomnia
Lasts <7 days Caused by a brief environmental or situational stressor (jet lag, overnight train journey, shift work) Sleep hygiene rules Hypnotics should be used at the lowest dose	**1 to 3 weeks** Caused by a personal stressor such as illness, grief or occupational problems Education about sleep hygiene Intermittent use of hypnotics- with the patient skipping a dose after every 1-2 days of good sleep	**>3 weeks** No specific stressor Underlying disease or personality disorder Non-Pharmacological measures Treatment of underlying illness Intermittent use of long acting hypnotic may be used

Non- Pharmacological Measures

- Education about sleep hygiene
- Reduce caffeine, tobacco and alcohol intake
- Reduce the fluid intake in the evening
- Avoid daytime napping
- Adequate exercise
- Relaxation training
- Take a glass of warm milk

* Effective in reducing sleep onset latency

NON-BENZODIAZEPINE HYPNOTICS

Zolpidem, Zopiclone, Zaleplon

- Also called “Z” drugs
- Not BZDs, acts on alpha 1 BZD site
- Produce sedative and hypnotic and weak anxiolytic action with negligible anticonvulsant, central muscle relaxant and amnesic action
- Least disruption of sleep architecture
- Least abuse potential
- Used for both transient and short term insomnia

- High safety margin
- Use of alcohol and other CNS depressants reduces its safety
- Overdose can be antagonized by Flumazenil(benzodiazepine competitive antagonist)

ZOLPIDEM

- Imidazopyridine derivative
- Rapid onset(30-60 mins)
- shorter action(t1/2= 2hrs)
- Equally effective hypnotic as BZDs – reduces the sleep latency and prolongs the total sleep duration
- Sedative action are strong than anxiolytic action
- Hypnotic effect is seen for a week without rebound insomnia after the withdrawal

- Readily absorbed orally, food reduces the absorption
- Bioavailability-70%
- Completely metabolized to inactive product
- Excreted via urine
- Dose- 5 to 10 mg at bedtime
- In elderly and hepatic dysfunction dose should be reduced to half
- Late night administration may cause delayed reaction time, morning sedation and anterograde amnesia

ZALEPLON

- Pyrazolopyrimidine derivative
- Effective in sleep onset insomnia
- Rapidly absorbed orally
- Shortest action(t1/2=1 hr)
- Bioavailability-30%
- Rapidly metabolised to inactive products
- Excretion- urine

- Sustained efficacy on prolonged use- no tolerance and no rebound insomnia
- Late night administration does not cause morning sedation, delayed reaction time, anterograde amnesia- can be taken by patient who is awake in the middle of the night and not able to sleep
- Dose- 5 -10 mg at bedtime

ZOPICLONE

- Cyclopyrolone derivative
- Short acting(t1/2= 5 hrs)
- Onset of action- 1 hr
- Useful for short term treatment of insomnia
- Prolongs stage 3 and 4 sleep but does not suppress REM sleep
- Less dependence liability, thus withdrawal is easier
- Side effects- metallic taste , reduced alertness and mental disturbances
- Dose- 7.5 mg before bedtime

ESZOPICLONE

- Active enantiomer of zopiclone
- Reduces sleep latency, number of awakenings, increase sleep time and helps in maintainance of sleep
- No after effects on next day
- Rapid oral absorption
- Bioavailability- 80%
- PPB- (50 to 60)%, longer action

- Used for both short term and long term treatment of insomnia for 1 year
- Tolerance and physical dependence is least
- Bitter taste
- Dose- 1-3 mg at bedtime

	ZOLPIDEM	ZALEPLON	ZOPICLONE
PLASMA HALF-LIFE	2hr	1hr	5hr
ALTERATION IN SLEEP PATTERN	Least	Least	May prolong SWS
PREFERRED USED	Sleep onset insomnia	Sleep onset insomnia	Short term insomnia
USE IN ELDERLY	Half dose		Half dose
NEXT DAY SEDATION	Infrequent	No	least
ADVERSE	Minimal	Not significant	significant

MELATONIN

- Hormone synthesized and released from pineal gland during night
- Maintains the circadian rhythm and sleep wakefulness cycle
- Reset the circadian rhythm which is disturbed in jet lag
- Used as chronobiotic
- Low dose- increases the tendency to fall asleep but no CNS depression
- High dose- produce hypnotic effects
- Dose- 3mg, two hour before bed time

Chronobiotic- agents that reset the biological clock or timing of circadian rhythm.

RAMELTEON

- 3 types of melatonin receptors- MT1, MT2, MT3
- MT1 and MT2- concerned with circadian rhythm
- MT3- related to intraocular tension
- Ramelteon is selective MT1 and MT2 receptor agonist- used for the treatment of insomnia
- Reduces sleep latency and improves sleep time
- Suitable for sleep onset insomnia
- More efficacious than melatonin but less efficacious than benzodiazepines

- Does not produce next day sedation
- No rebound insomnia, no addiction liability
- May lead to increased level of prolactin and decrease level of testosterone
- Adverse effects- headache, dizziness, fatigue, somnolence
- Dose- 8mg oral 30 mins before bedtime

OTHER SEDATIVE HYPNOTICS

Modafinil-

- Wake promoting agent
- Slow onset, duration of action is 8 to 12 hrs
- Low abuse liability
- Uses-
- For hypersomnias (narcolepsy or sleep apnoea syndrome)
- To maintain wakefulness in shift work sleep disorders

SUVOREXANT

- Orexin receptor antagonist
- Acts as a central promotor of wakefulness
- Hasten sleep onset, sleep maintainence and increase the total sleep duration
- Absorbed orally in 1 to 2 hours, t1/2=12 hrs, metabolised by CYP3A4, excreted in faeces
- Dose- 10 mg, 30 mins before bedtime
- Adverse effect- dose related
- Daytime somnolence. Impaired driving, unconscious night time behaviour

ATYPICAL ANXIOLYTICS

Buspirone, Ipsapirone, Gepirone

- Acts as a partial agonist primarily at brain 5HT1A receptors
- Not suitable in case of acute anxiety
- No muscle relaxant, anticonvulsant or hypnotic action
- Minimum abuse liability
- Less psychomotor skill impairment and does not potentiate the CNS depressant effects of alcohol
- Side effects- Tachycardia, nervousness, GIT distress

BETA ADREOCEPTOR ANTAGONISTS

Propranolol

- Used to reduce anxiety under conditions which provoke nervousness and anxiety
- Blocks peripheral manifestations of anxiety(palpitations, tremors)

Jane is a 35-year-old young woman who comes to your family practice clinic with the chief complaint that she has been abnormally tired, irritable, "stressed out" and tense the past 9 months, and these symptoms have not been getting any better. Her feelings of anxiety have been so difficult to control that she feels it is affecting her ability to deal with family and coworkers without "snapping" at them. She says that her constant struggle with anxiety is making her feel depressed as well. When asked questions about her past medical history, she does not appear to have had any prior issues with compulsive thoughts or behavior, or other forms of mental illness. She has not abused alcohol, or suffered from suicidal ideation. She has had difficulty falling asleep. After additional questioning, a provisional diagnosis of generalized anxiety disorder (GAD) is made, and you discuss beginning a treatment regimen that includes venlafaxine XR and a benzodiazepine. SNRI's have an onset of therapeutic effect that requires many weeks of continuous therapy. In contrast, for treating her anxiety disorder what is the expected onset of action of the benzodiazepine?

? immediate
? 1 week
? 3-6 weeks
? several months

Jane is prescribed venlafaxine XR (Effexor ®) 150 mg qd every morning, and lorazepam 0.5 mg tid. You warn Jane about drug side effects that include decreased balance & coordination & delayed reaction time (e.g. when driving). What other concern should you mention to her?

- ? increased insomnia
- ? increased risk of seizures
- ? intractable hiccups
- ? muscle stiffness
- ? respiratory depression if combined with alcohol

After 6 weeks of therapy, Jane is doing well, and is no longer plagued by anxiety. You recommend to Jane that she continue taking venlafaxine until her next follow-up visit, but explain that she can start a stepwise reduction in her dose of lorazepam over the next few weeks. You warn her to not to stop taking the drug all at once because that could cause unwanted & unnecessary withdrawal symptoms. Which of the following might Jane expect to experience if she did stop taking this medication "all at once" without a gradual tapering?

- ? anxiety & tremors
- ? hypotension
- ? nystagmus
- ? sedation

A 50-year-old man with a 30-year history of alcoholism and opioid addication is diagnosed with generalized anxiety disorder. Which of the following medications would be the best choice for treating his anxiety disorder?

- ? buspirone
- ? lorazepam
- ? midazolam
- ? zaleplon

A 42-year-old CEO comes to your family practice clinic complaining of insomnia and difficulty falling asleep. She has no symptoms of anxiety and admits having taken some triazolam (Halcion ®) provided by her secretary, but it made her drowsy and sluggish the following day. Which of the following could be used to treat her condition, has a relatively short duration of action, and targets a subset of benzodiazepine receptors expressing alpha-1 subunits?

? buspirone
? chlordiazepoxide
? fluoxetine
? phenobarbital
? zolpidem

THANK YOU